Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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1 Experience

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UNDP (ONU) Caracas, VE
Data Science Consultant *August 2019 - Present*

- ◦ Designed and develop workflows for topic modeling and community detection to analyze hundred-thousand-scale text and network data.
  ◦ Sped up the analytical workflow by 75% by redesigning the sampling strategy for jobs.
  ◦ Generated CLI tool for automation of predefined analysis workflows.

Technologies: Python, R, Bash, Gephi and Power BI.
Theory: Topic modeling, Community detection, Network Analysis and NLP.

- **Quash.ai** (remote) Caracas, VE
  Machine learning engineer *February 2019 - March 2020*
  ◦ Developed a customer facing API for using our internal credit scoring model.
  ◦ Improved credit scoring model performance of 5 business clients by 15% by incorporating state-of-the-art ML models adapted for our feature sets and hyper-parameter optimization models.

  Technologies: Python, R, S3 and Docker.
  Theory: Credit scoring and Hyper-parameter optimization.

- **Advoqt** (remote) Caracas, VE
  Machine learning engineer *February 2019 - August 2019*
  ◦ Implemented Deep learning model for classifying malicious urls using only urls' text features, achieving 90% of baseline accuracy by adapting state-of-the-art models on urls classifications.

  Technologies: Python, Elasticsearch and Spark.
  Theory: Deep learning and Text classification.

- **Universidad Central de Venezuela** Caracas, VE
  Advanced topics in AI(Adjunct Professor) *October 2018 - Present*
  ◦ Designed and organized course material and resources for topics described in here.
  ◦ Lectured material for students and grade their assignments.

  Introduction to Data Science (Teaching assistant) *April 2015 - February 2018*
  Data Mining (Teaching assistant) *August 2015 - February 2018*
  ◦ Designed and organized **practical** course material and resources for topics described in here.
  ◦ Lectured **practical** material for students and grade their assignments.

  Technologies: Python, Latex, R, Weka, Colab, Jupyter and swirl.
  Theory: Artificial intelligence, Data science and Data mining.

- **Fetcher** Caracas, VE
  Data Scientist *June 2018 - February 2019*
  - Estimated demand of required candidates for sourcing using daily output data.
  - Generated automated allocation algorithm for reducing overload of sourcing teams by 10% by using linear programming and constraint satisfaction techniques.

  Technologies: R, Shiny, Latex, SQL and Metabase.
  Theory: Linear programming and Constraint Satisfaction problems.

- **Predictvia** Caracas, VE
  Artificial Intelligence *July 2016 - January 2017*
  Artificial Intelligence intern *February 2016 - July 2016*
  - Performed social network analysis, exploratory data analysis, supervised and unsupervised learning
  - Built and implemented classification models, perform feature engineering and calibrated existing models.

  Technologies: Python. **Theory:** Social network analysis, Attribution models and Recommendation systems.

2 Education

[leftmargin=*]
Universidad Central de Venezuela Caracas, Venezuela
B.Sc in Computer Science *Sept. 2012 – Jun. 2018*

- Took elective courses on: Introduction to data science, Advanced topics in data science, Artificial Intelligence, Data mining, Text mining, NoSQL databases, Digital image processing.
- (Thesis title) AdaBnn: Binarized Neural Networks trained with adaptive structural learning.

Following sections items are clickable

3 Certifications

[leftmargin=*]
Coursera MOOC
R Programming - ID: MZ7TC7ADV7YY *Issued September 2016*

Text Mining and Analytics - ID: 36JC577GBFYD *Issued August 2016*
Text retrieval and search engines - ID: MDWATFKLPH9Z *Issued June 2016*
The Data Scientist's Toolbox - ID: TM7T8CJXNV6R *Issued June 2016*
Data Visualization - ID: WYGG7TDFPB3F *Issued March 2016*

4 Projects

[leftmargin=*] AdaBnn: Binarized Neural Networks trained with adaptive structural learning. WarmChat: This app is a small chat room that classify every message and say weather is an insult or not using four classifiers: svm, random forest, decision trees and max entropy. Small tool for generating frequent words analysis on the position names of LinkedIn Connections. Template for Dockerized Shiny (R) application deployment.

5 Personal publications

[leftmargin=*] Sajidxa Mariño, Jesus Lares, Wilmer Gonzalez. Diode's Laser in Office Endoscopic Surgery Center for Nasal Obstruction and Snoring. J Med - Clin Res & Rev. 2018; 2(6): 1-3. Twitter, A data source for the academy (spanish). GitHub & Social Coding. Social network for programmers (spanish). Reproducible Research. Case of study: allocation of university places in UCV by OPSU (spanish).

6 Other programming tools

[leftmargin=*] **Cloud services:** AWS(Lambda, S3, Sagemaker), Azure(Blob storage). **Deep learning frameworks:** Keras, Tensorflow. **R:** dplyr, ggplot, tidyr, shiny, plumber. **Python:** xgboost, sci-kit, pandas, falcon, flask. **Webserver:** Gunicorn.